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# Experimental research on aero-propulsion coupling characteristics of a distributed electric propulsion aircraft

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**KEYWORDS**

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Boundary layer ingestion;
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Wind-tunnel experiment

**Abstract** Distributed Electric Propulsion (DEP) aircraft use multiple electric motors to drive the propulsors, which gives potential benefits to aerodynamic-propulsion interaction. To investigate and quantify the aerodynamic-propulsion interaction effect of the wing section, we built a DEP demonstrator with 24 "high-lift" Electric Ducted Fans (EDFs) distributed along the wing's trailing edge. This paper explores and compares the aero-propulsion coupling characteristics under various upstream speed, throttle, and EDF mounting surface deflection angles using a series of wind tunnel tests. We compare various lift-augmentation power conditions to the clean configuration without propulsion unit under the experiment condition of 15–25 m/s freestream flow and angles of attack from −4° to 16°. The comparison of computational results to the experimental results verifies the effectiveness of the computational fluid dynamic analysis method and the modeling method for the DEP configuration. The results show that the EDFs can produce significant lift increment and drag reduction simultaneously, which is accordant with the potential benefit of Boundary Layer Ingestion (BLI) at low airspeed.
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## 1. Introduction

To reduce the environmental impact of aviation activities, industries and research institutions are taking necessary steps to minimize aviation exhaust emissions. In recent years, both NASA's New Generation (N + 3) subsonic aircraft performance target and the EU's "Flightpath 2050" Plan set a much higher requirement than the existing standards for future aircraft fuel consumption, exhaust emission, and noise level.[1]

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NASA envisions electrically powered airplanes filling the skies and eliminating the use of jet fuel that pollutes the atmosphere with carbon dioxide, nitrogen oxides, hydrocarbons, and soot particles.[2,3]

Distributed Electric Propulsion (DEP) technology uses electricity as an energy source to drive geometrically distributed propellers or ducted fans, utilizing potential aerodynamic-propulsion interaction benefits.[4,5] The propulsion unit installed near wing surfaces can provide high energy flow to increase the lift by introducing a high energy air stream to support a pressure load beyond the airfoil trailing edge.[6] Thus, the aircraft with DEP systems allows a smaller wing (i.e., shorter chord and higher aspect ratio), with lower cruise drag, and only consumes a fifth of energy, compared with a traditional gas-fueled airplane.[7,8] Many of the recent Vertical/Short Take-Off and Landing (V/STOL) aircraft selected the fixed-wing configuration with DEP to archive a high-efficiency cruise capability. During the transition from hover to forwarding flight, the traditional control surfaces cannot provide enough controllability until the aircraft's stall speed is reached. Therefore, it is necessary to investigate the aero-propulsion coupling characteristic of the DEP systems and explore the benefit of increasing lift and improving vehicle controllability.

Limited by current low-density electric storage technology, recent research mainly focuses on general aviation. However, as the batteries and MW-level electric motors continually develop, the DEP technology can be used in a commercial airliner in the future. Recent cases include NASA's X-57 Maxwell test aircraft[9,10] and German Lilium Aviation's Lilium Jet.[11] The researchers at NASA investigated DEP technology in the Leading-Edge Asynchronous Propulsion Technology (LEAPTech) project and designed the experimental aircraft X-57, which replaced the original wing and internal combustion engines of a Tecnam P2006T aircraft with a redesigned smaller wing with an electric-powered high lift system and all-electric propulsion system.[12,13]

Inspired by former researchers' work, we designed a distributed electric propulsion STOL technology verification UAV with a twin-fuselage tandem wing configuration. The aircraft's configuration and parameters are shown in Fig. 1 and Table 1.

The tandem configuration has been widely used in the design of DEP S/VTOL aircraft, such as the Lilium Jet, which proves that the advantage of this configuration in terms of the position of the center of gravity contributes to the VTOL of aircraft based on the tilt-wing or tilt propulsion. In addition, the tandem configuration has the advantage of deploying the DEP system. For DEP aircraft, the thrust of the EDFs is distributed spanwise along the wing's trailing edge and needs to transfer to the aircraft structure, which increases the requirements for the strength and stiffness of the wing structure, resulting in increased wing mass. The twin-fuselage configuration replaces the centrally positioned fuselage, which reduces the wing bending moment and allows a lighter wing structure.[14]

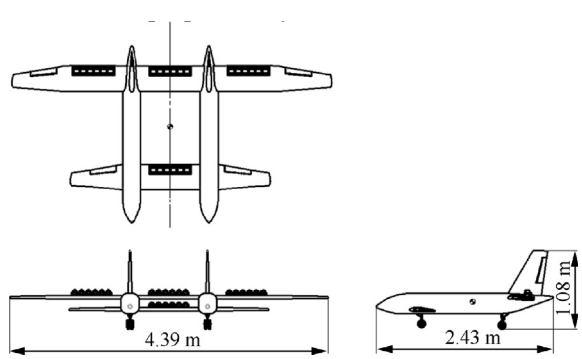



**Fig. 1** Configuration of DEP STOL technology verification UAV.

**Table 1** Design parameters of DEP STOL technology verification UAV.

| Design parameter | Value | |
|---|---|---|
| | Front wing | Rear wing |
| Mass (kg) | 40 | |
| Length (m) | 2.43 | |
| Span (m) | 4.39 | |
| Height (m) | 1.08 | |
| Cruise speed (m/s) | 25 | |
| Stall speed (m/s) | 15 | |
| Airfoil | NACA 4415 | |
| Wing area ($m^2$) | 0.804 | 1.368 |
| Wing span (m) | 2.74 | 4.388 |
| Aspect ratio | 8.93 | 13.60 |
| Taper ratio | 0.5 | 0.5 |
| Root chord (m) | 0.345 | 0.345 |
| Tip chord (m) | 0.173 | 0.173 |
| MAC (m) | 0.311 | 0.325 |

The power unit configuration of the verification UAV is different from NASA's X-57's leading-edge propeller. Using Electric Ducted Fans (EDFs) spanwise distributed along the wing's trailing edge, could accelerate the boundary-layer flow on the wing's upper surface, increasing the lift and reducing the drag.[15] Besides, the downwash of the front wing can reduce the local Angle of Attack ($\alpha$) of the rear wing. Therefore, when the front wing stalls, the rear wing can stay below the stall angle of attack, which can delay the stall and increase the maximum lift coefficient of the aircraft.

The whole UAV is equipped with twenty-four EDFs divided into four sets of power units. Each one includes six EDFs propulsion units, as shown in Fig. 2. The EDFs mounted trailing-edge portion was connected to the wing through a shaft, and it tilted as the shaft rotated, driven by the servo system mounted in the wing. The EDFs, together with the trailing edge portion of the wing with the EDFs mounted, were tilted between horizontal and vertical configurations to accomplish horizontal flight and STOL.[16]

Despite the large amount of ongoing research related to DEP aircraft, especially for the aerodynamic-propulsion interaction, the result varied due to different configurations.[17,18] We conducted a series of ground test experiments and test flight experiments to explore the characteristics of aero-propulsion coupling.[19] A non-command nose-up occurred during the flying test when the pilot tried to slow down and perform the flare maneuver to land the aircraft. Under particular circumstances, when the pilot reduces the thrust, the angle of attack can rapidly raise with a speed up to 6 (°)/

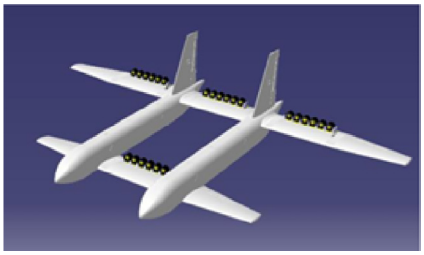


**Fig. 2** Configuration of DEP technology verification UAV.

s. It is a hazardous phenomenon that can easily stall the aircraft.

Furthermore, the ground test result shows that the maximum lift at a small angle of attack can be increased by 114%. At the same time, the aerodynamic center of the aircraft moves nearly 30 cm to the back, creating a noticeable nose-down moment.

To this end, the aero-propulsion coupling characteristic of a tandem wing DEP UAV is studied; especially, the impact of throttle and propulsion installation angle of EDFs on the aerodynamic performance under various angle of attack is explored. A series of wind tunnel experiments of the wing section and CFD analysis were designed and conducted to better explain the non-command nose-up phenomenon and unveil the aero-propulsion coupling characteristic of DEP aircraft.[20] Both lift and drag coefficients are sensitive to the thrust change, and the aero-propulsion coupling characteristic is quantified. Furthermore, a numerical method is developed and verified, which provides a rapid assessment tool for evaluating the aero-propulsion coupling characteristic of DEP UAVs.

## 2. Wind-tunnel experiment scheme

### 2.1. Model design and construction

As shown in Fig. 3, the wind tunnel model was configured as a NACA 4415 airfoil with a deflectable power unit with a 345 mm chord and 867 mm span. The deflectable surface was equipped with six EDFs mounted on the installation-angle-adjustable fixed stand on the upper surface of trailing edge. The estimated aerodynamic parameters of powered wing section were shown in Table 2.

The STOL aircraft we designed could utilize the deflectable trailing edge flap, which the DEP system was mounted on, to generate a vertical thrust vector during the short takeoff and convertion period. Therefore, the performance of EDF was slightly larger than that found from a typical fixed-wing DEP system designed for conventional takeoff and landing operations and cruising. Among the commercially on-shelf EDFs that met the performance needs, the FMS technologies 12-blade 70 mm EDF driven by a 3060-KV1900 25.2 VDC motor with 80 A Electronic Speed Controllers (ESCs) was selected. A 15 kVA (DC30 V/500 A) programmable high-power regulated DC power supply is used according to EDF's rated voltage and power demand. Moreover, the throttle of the EDF was controlled by a Pulse-Width Modulated (PWM) electric motor controller.

**Table 2** Estimated aerodynamic parameters of powered wing section.

| Parameter | Value |
|---|---|
| Maximum lift coefficient | 1.8 |
| Minimum drag coefficient | 0.033 |
| Cruise lift to drag ratio | 12 |

All major structural components of the model were 3D printed using a Selective Laser Sintering (SLS) process. The deflectable surface with the power unit was installed on the wing section with a predesigned deflection angle of 0°/15°/30°. Two connector parts made of SUS304 stainless steel connected the wing to the 6-component force balance sensor, as shown in Fig. 4.

### 2.2. Wind tunnel experimental device

The experiment was conducted in a low-speed wind tunnel with a 1.5 m × 1.5 m test section at Northwestern Polytechnical University. This wind tunnel features a maximum test-section velocity of 45 m/s and a freestream turbulence intensity of Tu < 0.2% for all operating speeds.

The 6-component force balance calculates the three-axles force and moment coefficient. These forces and moments were translated into wind axes using a standard coordinate transform by the airfoil angle of attack to obtain the net streamwise and stream-normal force, as shown in Fig. 5(a).

The balance was installed on an angle of attack adjustable tail support system, as shown in Fig. 5(b). The angle of attack adjustable range is from −10° to 110° with an error of less than ±1′.

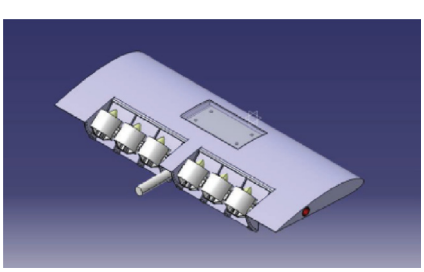
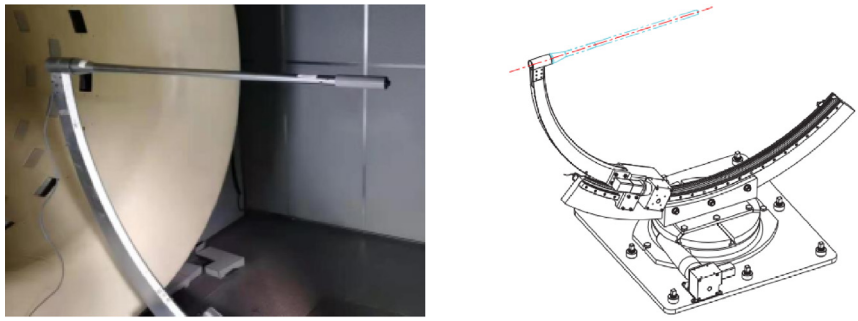


**Fig. 3** CAD model of experimental wing section.

**Fig. 4** Wind tunnel experimental device and wing section assembly.



(a) 6-component force balance assembly (b) Angle of attack adjustable tail support system

**Fig. 5** Wind tunnel experimental device.

### 2.3. Experimental parameters

The parameters measured were upstream airspeed ($V$), angle of attack ($\alpha$), and throttle ($T$), defined as the percentage of the full-throttle PWM signal sent to the speed controllers, and the deflection angle of the EDF-mounted surface ($\delta_T$). A static thrust test was conducted at the beginning of the test to establish the relationship between PWM pulse width ($P$) and the thrust of EDFs. Thus, it can help to eliminate the influence of thrust on aerodynamic force measurement and define the CFD calculation condition. Table 3 lists the tested values for all parameters.

### 2.4. Experimental data processing

All the data measured in this experiment are functions of the angle of attack ($\alpha$), throttle ($T$), and deflection angle ($\delta_T$). The measured force and moment are in the balance axis coordinate. The definition of the balance axis system was: taking the balance calibration center as the coordinate origin $O$, the inward direction along the balance axis is the positive direction of the $x$ axis, the downward direction perpendicular to the balance axis is the positive direction of the $y$ axis, and the positive direction of the $z$ axis is perpendicular to the $xOy$ plane to form the right-hand system.[21] Therefore, to obtain the lift ($L$) and drag ($D$), the coordinate transformation is required when processing the measured value of $x$ axis force ($F_x$) and $y$ axis force ($F_y$), as shown in Fig. 6 and Eq. (1);

$$\begin{cases} L = -F_x \sin\alpha - F_y \cos\alpha \\ D = F_x \cos\alpha - F_y \sin\alpha \end{cases} \tag{1}$$

In addition, the torque ($M$) is measured by taking the balance calibration center as the reference point, the torque at 25% MAC position ($M_{25\%\mathrm{MAC}}$) can be obtained according to Eq. (2):

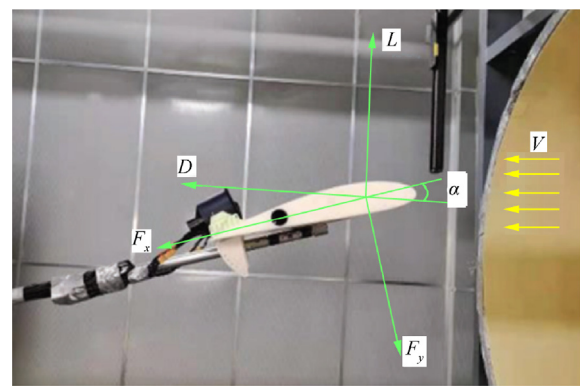



**Fig. 6** Balance axis coordinate system.

$$M_{25\%\mathrm{MAC}} = M - Ll_1 - Dl_2 \tag{2}$$

where $l_1$ and $l_2$ refer to the horizontal and vertical distance between the 25%MAC and balance calibration center, respectively.

## 3. CFD simulation scheme

### 3.1. Governing equation

For the STOL DEP UAV mentioned above, due to the low flight speed between 20 m/s and 30 m/s, the three-dimensional incompressible Navier-Stokes (N-S) equations are used as the governing equations, which in Cartesian coordinates can be written as Eqs. (3)–(6):

$$\nabla \cdot \vec{V} = 0 \tag{3}$$

$$\rho \frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \mu \nabla^2 u + \rho f_x \tag{4}$$

**Table 3** Test parameters and values in various airspeed cases.

| Parameter | Value | | | |
|---|---|---|---|---|
| | $V=$ 0 m/s | $V=$ 25 m/s | $V=$ 20 m/s | $V=$ 15 m/s |
| Angle of attack $\alpha$ (°) | −4 : 2 : 18 | | 0 : 4 : 16 | |
| EDF-mounted flap deflection angle $\delta_T$ (°) | 0/15/30 | | 0 | |
| Throttle $T$ (%) | 0-100, Segment:10 | | | |

$$\rho \frac{Dv}{Dt} = -\frac{\partial p}{\partial y} + \mu \nabla^2 v + \rho f_y \tag{5}$$

$$\rho \frac{Dw}{Dt} = -\frac{\partial p}{\partial z} + \mu \nabla^2 w + \rho f_z \tag{6}$$

where $\rho$ and $p$ denote density and static pressure, respectively, $u$, $v$, $w$ denote Cartesian velocity components, $\mu$ denotes viscosity coefficient, and $f_x$,$f_y$, $f_z$ denote the body force components per unit volume.

The flow solver is an in-house incompressible Navier-Stokes code. In this code, the Semi-Implicit Method for Pressure-Linked Equations (SIMPLE) algorithm is used for the pressure–velocity coupled equation. The working fluid is assumed to be ideal gas, and turbulence is modeled using the Spalart-Allmaras (S-A) model, which is suitable for engineering calculation. The 2nd-order centered scheme is used as a pressure discretization scheme, while the 2nd-order upwind scheme is used for momentum equations.

### 3.2. Rotor disk model and computational grid

The configuration of a single fan is complex enough, mainly because it has a cylindrical duct and 12 rotor blades. When the motor, cone, and mounting bracket structure are neglected, as seen in Fig. 7(a), it is still too complicated for numerical simulation. If the rotor blades are individually modeled, they will be too expensive, both in computational expense and time needed to generate grids. In order to simulate EDF's influence on wing performance quickly, the rotor disk method is adopted. This method ignores the geometric shape of the fan rotor blade and regards it as a disc without thickness. The simplified ducted fan is shown in Fig. 7(b).

The fan rotor is simulated by imposing a boundary condition at the surface swept by the fan rotor. Across this surface of the disk, there is a pressure jump ($\Delta p$), as shown in Eq. (7):

$$\Delta p = p_2 - p_1 = \frac{F_T}{A} \tag{7}$$

where $p_1$ and $p_2$ denote the pressure above and below the rotor disk, respectively, $F_T$ is the fan's thrust, and $A$ is the disk area.

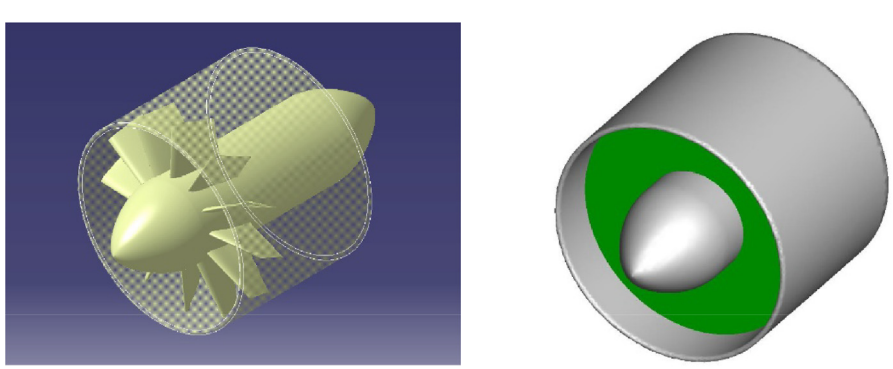
Considering that the wing section is a straight rectangular wing, and the ducted fans are symmetrically distributed, the half model is used for calculation. A semi-cylindrical region is selected as the calculation domain. To ensure that boundary location does not influence the flow, the domain boundary upstream and downstream are placed at a distance of 20 times the half wingspan length $b$, while the calculation domain radius is also 20 times the half wingspan length $b$. The calculation domain is shown in Fig. 8.

Because the power unit at the wing's trailing edge is multiple ducted fans, the unstructured grid is used to discretize the calculation domain to reduce the difficulty of mesh generation. Triangle is used to discretize the surface and let grid points cluster at places with high curvatures, such as leading-edge, trailing-edge, and duct inlet. In order to accurately simulate the flow inside the boundary layer, prims are employed to discretize the boundary layers. The height of the first grid layer is $1 \times 10^{-5}$ chord of the wing. The grid layer height increases uniformly from the wall with a growth rate of 1.2, and the total number of layers is 30. The rest of the flow domain is then filled with tetrahedra. And 7.79 million grid cells in total are generated for simulation. A grid independence study was conducted by refining the grid on wall surfaces and in both boundary layers. No considerable changes were observed when the grid was refined, and all simulations were performed on the grid above. The computational grids of rotor disk, ducted fan, and wing surface are shown in Fig. 9.

## 4. Results analysis and discussion

### 4.1. Baseline test result

To quantify the thrust performance of the system, and specify the relationship between the PWM control signal and EDFs thrust, force balance measurements were initially taken for each case under static conditions. Due to the installation angle of the EDF, the thrust vector is not aligned with the wing's chord. The thrust vector was measured by the 6-component force balance in the chord's normal and axial directions. The EDF's throttle was defined as the pulse width percentage of the full-throttle PWM signal sent to the electric motor speed controllers. The data presented in this section were acquired as all six of the EDFs operate at the same throttle. The 0% throttle case corresponds to a fan not receiving power. The EDF's RPM is unknown due to the difficulty of installing the RPM sensor without the influence on the aerodynamic performance.



(a) Geometry model of EDF (b) Simplified electric ducted fan model

**Fig. 7** Modeling of electric ducted fan.

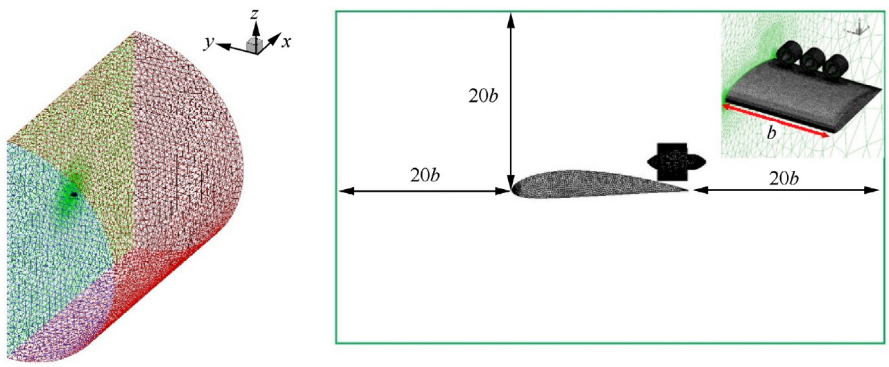
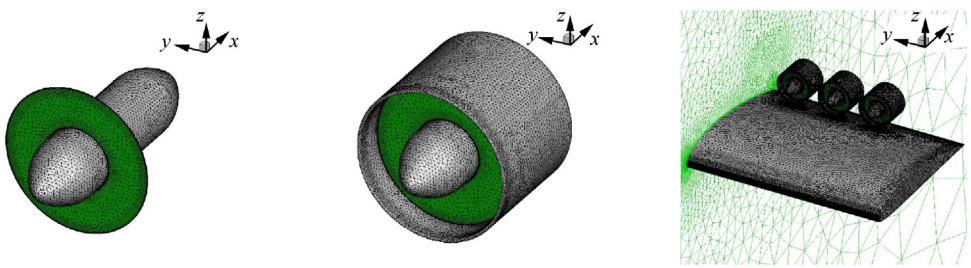



**Fig. 8** Schematic of calculation domain.




(a) Grid of rotor disk (b) Grid of electric ducted fan (c) Grid of whole model

**Fig. 9** Schematic of computational grids.

The thrust under various freestream airspeeds is shown in Fig. 10(a). The result shows that the total thrust declined as the airspeed increased. The thrust reduced to 53.7 N under $V = 25$ m/s compared to 93 N under $V = 15$ m/s. The static thrust and the PWM input signal could be considered a linear relationship despite slight deviations around 70% throttle in the situation of no EDF-mounted surface deflection angle. The total thrust of 6 EDFs could reach 88.7 N.

Besides, to obtain the primary aerodynamic characteristic of the wing section, a clean wing configuration test with various EDF-mounted surface deflection angles was conducted. The result was used as the cross-reference and primary aerodynamic performance for the powered model experiment, as shown in Figs. 10(b)–(c). The result shows that the lift-to-drag ratio at cruising angle of attack of the clean configuration wing section is 7.2, and the maximum lift coefficient ($C_L$) can reach 0.91 at $\alpha = 16°$.

### *4.2. Throttle change effect on aerodynamic performance*

The lift and drag coefficients were obtained under various angle of attack and throttle to quantify the propulsion system's effect on aerodynamic performance.

$C_L$ versus $\alpha$ as a function of $T$ is shown in Fig. 11. Compared to the clean configuration, $C_L$ significantly dropped for $T$ between 0% and 20% under every experiment airspeed. For the DEP aircraft, when the power unit mounted on the upper wing surface was working at a windmill or in idle situations, the airflow over the power unit would separate from the wing surface, causing the lift loss. As observed in Fig. 12, where $C_L$ versus $T$ is presented as a function of $\alpha$, there is an apparent increase in $C_L$ for a given $\alpha$ as the $T$ increases. Compared to clean configuration, there is also a slight $C_L$ increase around $T = 60\%$–$70\%$ while $\Delta C_L$ can reach 0.30 for $V = 20$ m/s and 0.25 for $V = 25$ m/s.

Fig. 13 and Fig. 14 show the model's drag coefficient ($C_D$) versus $\alpha$ and $T$. The first trend to be noticed is that the additional drag from the power unit is significant for each case. The powered wing section drag is increased compared with the clean configuration under $T = 0°$–$20\%$. As observed in Fig. 14, the installation of the power unit caused a significant drag. In the $V = 15$ m/s and 20 m/s cases, the increased $C_D$ can be reduced at all angle of attack as the throttle increases, especially from 70% to 100%. However, the reduced $C_D$ is still slightly larger than the clean configuration. The same trend can be found under $V = 25$ m/s. For results at $\alpha = 0°$–$12°$, the increased $C_D$ can be reduced to a slightly lower level than the clean configuration in a high throttle state. The lift to drag ratio at cruising angle of attack can reach 21.57 as the $T = 100\%$. It shows the possibility to reduce the drag while keeping the lift unchanged by reducing the wing area.

The impact of the throttle and propulsion unit installation angle of EDFs on the aerodynamic performance at various angle of attack is noticeable. The $C_L$ loss in the no power situation was about 0.25 for $V = 20$ m/s and 25 m/s, while $C_D$ increased up to 0.3, which resulted in the non-command nose-up mentioned above. The flare maneuver requires reducing the throttle, which could reduce the lift and drastically increase drag according to the result.

When designing the DEP aircraft power system, we need to consider that, to avoid the lift loss, the minimal throttle should be analyzed and determined through experiment and CFD. At least 20% of thrust was needed to cover the $C_L$ loss in our case.

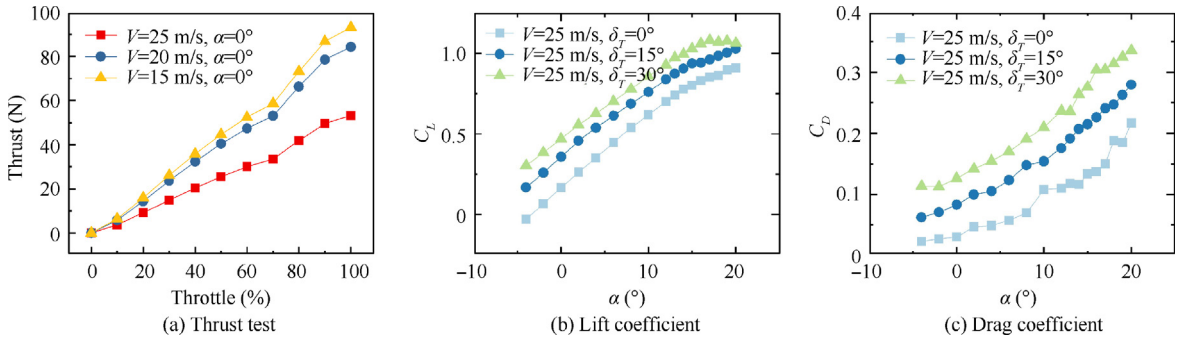



**Fig. 10** Static thrust baseline test results at different EDF-mounted surface deflection angles.

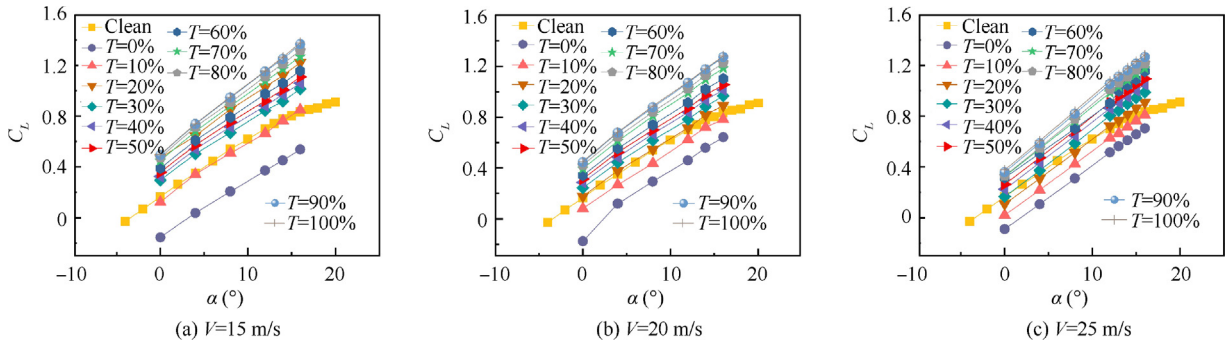



**Fig. 11** Lift coefficient versus angle of attack under various airspeed.

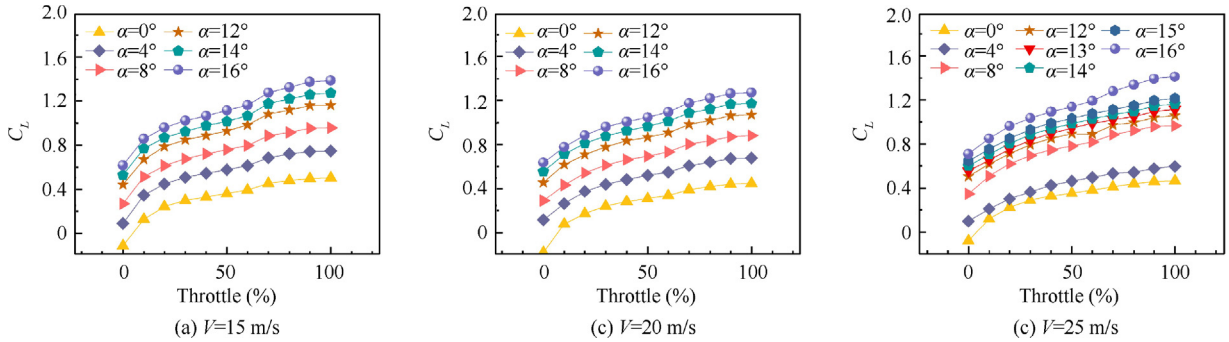



**Fig. 12** Lift coefficient versus throttle under various airspeed.

These results also show the potential benefit of a larger $\Delta C_L$ that could be obtained at high angle of attack for STOL vehicles with powered high-lift systems. It is interesting to note that $C_L$ increment is roughly the same per 10% throttle. Besides, with the help of the EDFs intake, the stall delays to a higher angle of attack compared to the clean configuration.

While distributed propulsion-based control is usually designed to provide yaw authority, the changes in $C_L$ associated with throttle variation highlight the potential to utilize distributed propulsion as a roll controller and might be helpful for the attitude control during the takeoff/landing of STOL vehicle. Furthermore, studies on EDF inlet and nozzle optimization and EDF- airfoil combination design are needed to improve the aerodynamic efficiency.

### *4.3. Effect of EDF installation surface's deflection angle*

Data were collected at various EDF-mounted surface deflection angle to assess their influences on the propulsion system. It should be noted that traditional lift and drag coefficient definitions do not account for the effects of thrust from the DEP system on stream-normal and stream-wise force coefficients.

With increasing $\delta_T$, the net thrust increase of the DEP system is noticeable, as shown in Fig. 15. When the $\delta_T$ = 15°–30°, it was evident that the chord normal direction thrust increased by 30–50 N, with almost unchanged chord axial direction thrust. With deflection of the surface, the wing's block effect on EDF's inlet lips was reduced, and a noticeable 16 N (~30%) total thrust augment was found. This phenomenon

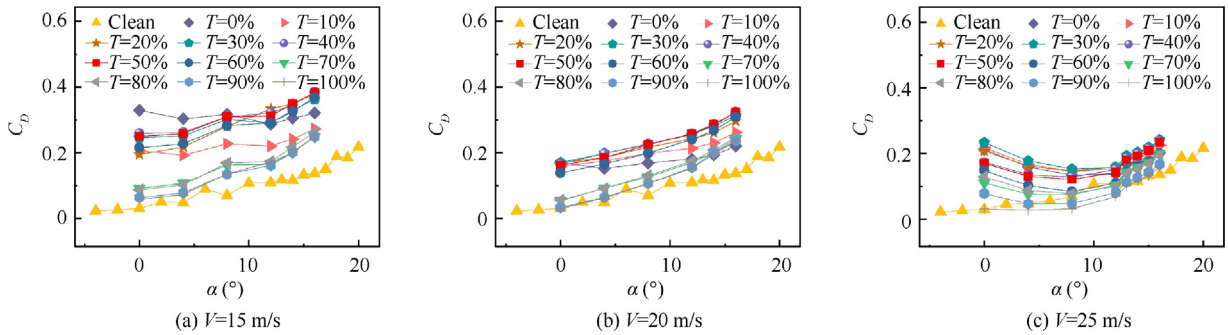



**Fig. 13** Drag coefficient versus angle of attack under various airspeed.

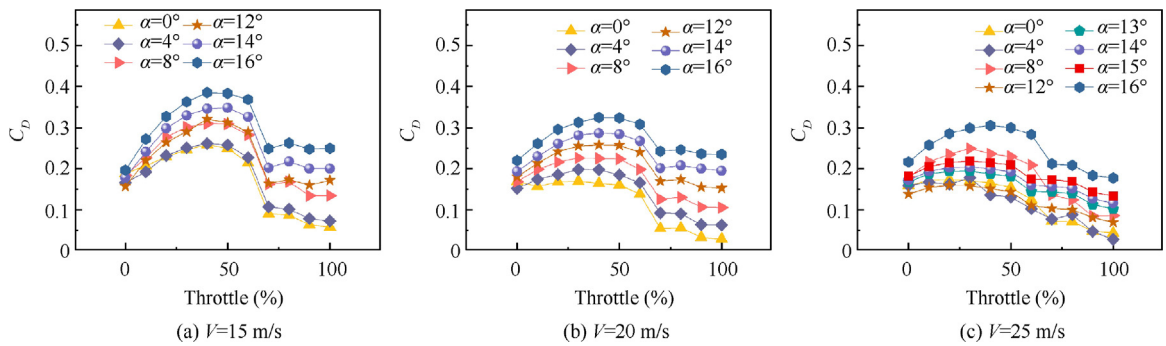



**Fig. 14** Drag coefficient versus throttle under various airspeed.

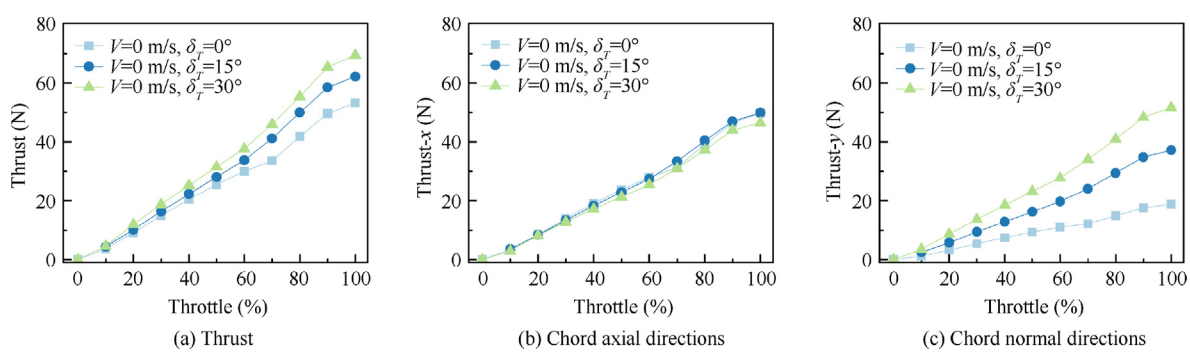



**Fig. 15** Baseline test results at various EDF-mounted surface deflection angle.

should be carefully studied to benefit the EDP propulsion system design and STOL characteristics.

The lift coefficient versus angle of attack and versus throttle under various EDF-mounted surface deflection angle are shown in Fig. 16 and Fig. 17, respectively. The observed overall trends were an expected increase in $C_L$ due to increased $V$ and $\delta_T$. The $C_L$ reached 2.28 at $V$ = 25 m/s, $\alpha$ = 16° and $\delta_T$ = 30°, compared to 1.429 when $\delta_T$ = 0°. $C_L$ above 20% throttle at $\delta_T$ = 15° and 30° showed a linearly increasing relationship with the throttle, exceeding $C_L$ of clean configuration. The deflection of EDFs mounting surface's improvement to $C_L$ was noticeable and can benefit the STOL performance of the DEP aircraft.

Meanwhile, the same trend of $C_D$ change is shown in Fig. 18 and Fig. 19. The thrust of different surface deflection angles was measured, and the $C_D$ increment caused by the thrust device under a small angle of attack was eliminated. Since the thrust increases and $C_D$ reduces with the slight $\delta_T$, and it could benefit the cruising performance of the DEP aircraft, that is, adding a slight thrust installation angle to improve aircraft performance. Besides, the gradient of $C_D$ to $T$ and $C_D$ to $\alpha$ increased as the $\delta_T$ increased.

### 4.4. Verification of numerical method

In order to verify the reliability of the numerical method used, the clean wing section and the wing section with power unit are calculated respectively to compare with experimental data. The clean wing section refers to the removal of the power unit. Fig. 20 shows the difference between it and the powered

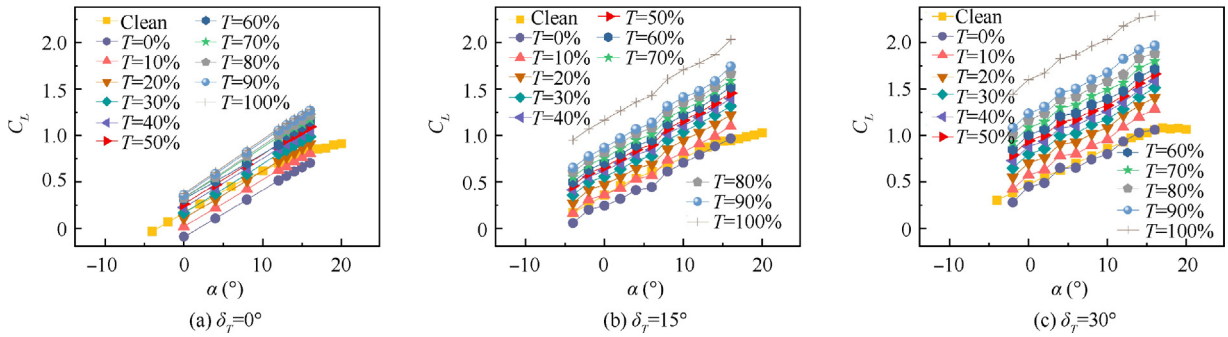



**Fig. 16** Lift coefficient versus angle of attack at various EDF-mounted surface deflection angle.

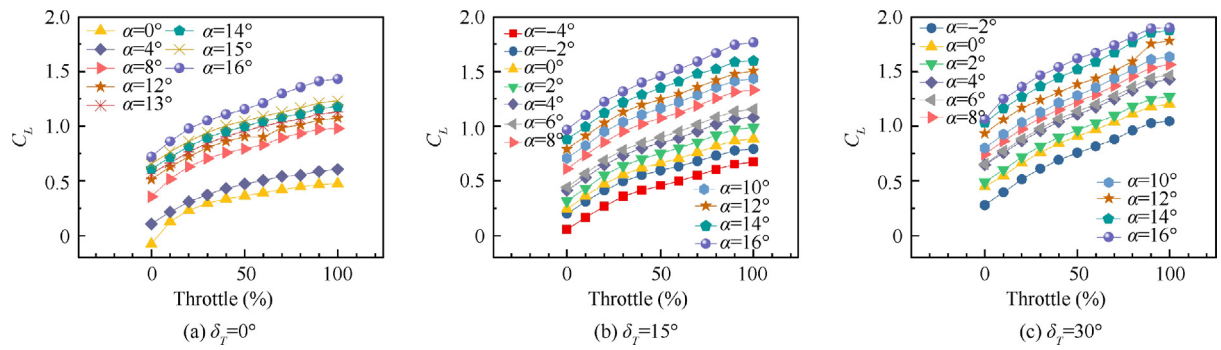



**Fig. 17** Lift coefficient versus throttle at various EDF-mounted surface deflection angle.

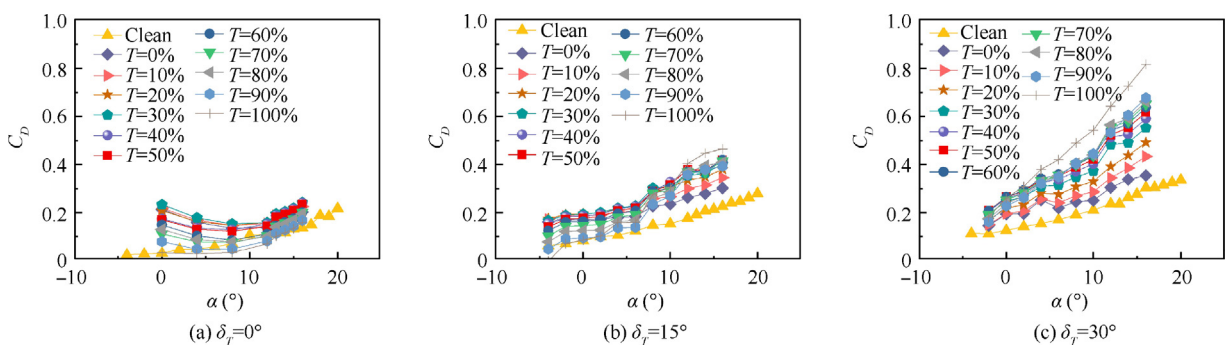



**Fig. 18** Drag coefficient versus angle of attack at various EDF-mounted surface deflection angle.

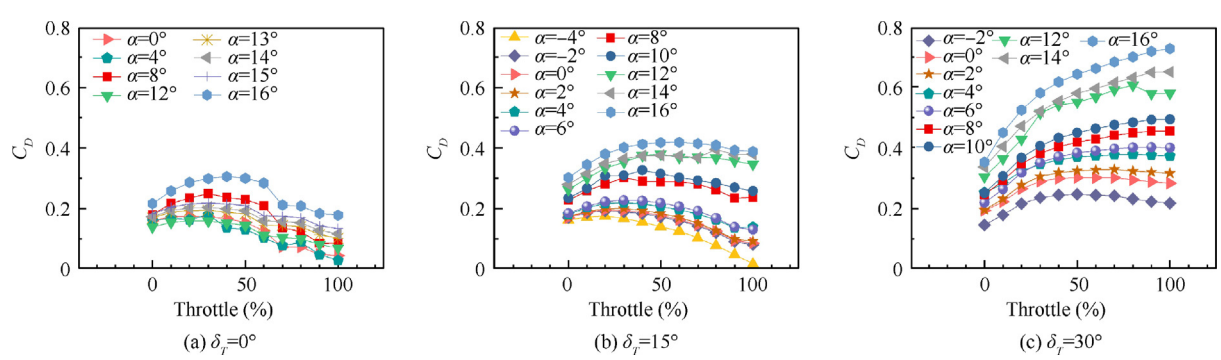



**Fig. 19** Drag coefficient versus throttle at various EDF-mounted surface deflection angle.

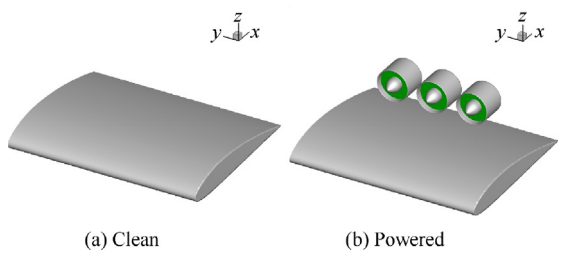



**Fig. 20** Model comparison of clean wing to powered wing.

**Table 4** Calculation status of clean wing section.

| Parameter | Value |
|---|---|
| Airspeed (m/s) | 25 |
| Atmosphere pressure (kPa) | 101.325 |
| Temperature (K) | 288 |
| Angel of attack (°) | 0:4:20 |

**Table 5** Calculation status of wing section with power unit.

| Parameter | Value |
|---|---|
| Airspeed (m/s) | 25 |
| Atmosphere pressure (kPa) | 101.325 |
| Temperature (K) | 288 |
| Angel of attack (°) | 0 |
| Thrust (%) | 0:10:100 |

model. The calculation status of the clean wing section and wing section with the power unit is shown in Table 4 and Table 5, respectively.

In Fig. 21, the calculated lift and drag of the clean wing section at different angles of attack are plotted in comparison with the experimental data. It can be seen clearly that the lift and drag increase as the attack angle increases. The calculation results are in good agreement with the experimental data. Both values and trends are accurately predicted.

In Fig. 22, the calculated lift and drag of powered wing sections under different throttle are plotted compared to the experimental data. It is clearly seen that the lift increases as the throttle increases, but the drag decreases as the throttle increases. The calculated lift agrees well with the experimental data. The accuracy is lacking for calculated drag, but the trend is well predicted. The lack of accuracy is mainly due to the complex nonlinear flow of the EDF and the simplified CFD model to reduce the expense of simulation. We consider that the numerical method is still reliable and can be used as a rapid evaluation tool for the aerodynamic performance of the wing with power unit.

## 5. Conclusions

A hazardous scenario that inspired this research occurred during the final approach stage of the DEP UAV flying test. When the pilot reduces the thrust, the angle of attack could rapidly raise with the angular speed up to 6 (°)/s and lead to a stall. The steady, aero-propulsion coupling characteristics and vector thrust characteristics for a distributed electric propulsion system were studied to explore the underlying mechanism. The aero-propulsion characteristics and performance of a distributed ducted fan propulsion system were experimentally investigated at a 1.5 m × 1.5 m low-speed wind tunnel owned by Northwestern Polytechnical University. The stream-wise and stream-normal aerodynamic and propulsion force data were collected as a function of $\alpha$ and $T$ to quantify the aero-propulsion performance of the DEP UAV. Meanwhile, it is essential to explore the trends and underlying mechanisms for the STOL vehicle with a vector distributed propulsion system.

The experiment verified the BLI's effectiveness of the DEP system. With the help of the DEP propulsion system, the $L/D$ at cruising angle of attack $\alpha = 4°$ can be increased from 7.2 to 21.57 (full throttle) while the max $C_L$ can be increased from 0.836 to 1.429 at $\alpha = 16°$ (full throttle). However, while $C_L$ increases, $C_D$ is drastically increased due to the additional structure of EDFs. These findings explained the non-command nose-up well. When the pilot tries to land the UAV, a decrease of throttle is required, which results in the decreased lift and increased drag. The rear wing area of the UAV is 170% of the front wing with three propulsion units,

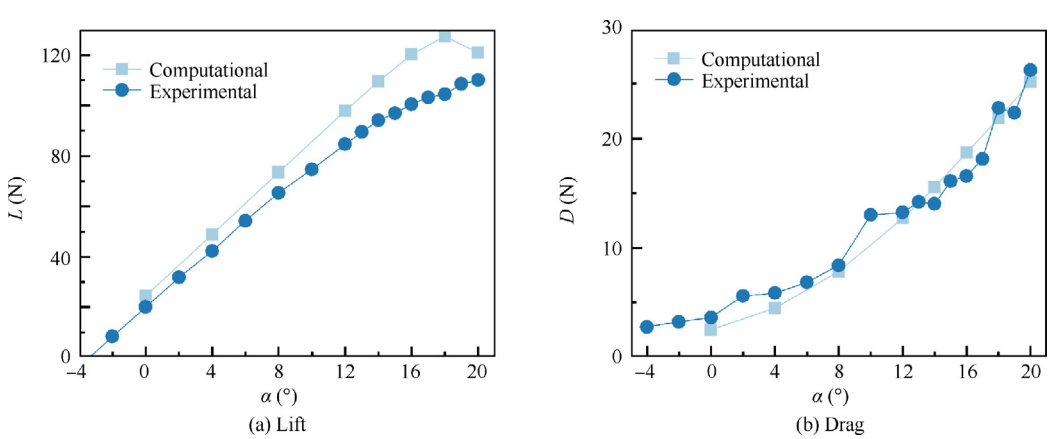



**Fig. 21** Comparison of CFD results of clean wing to experimental results.

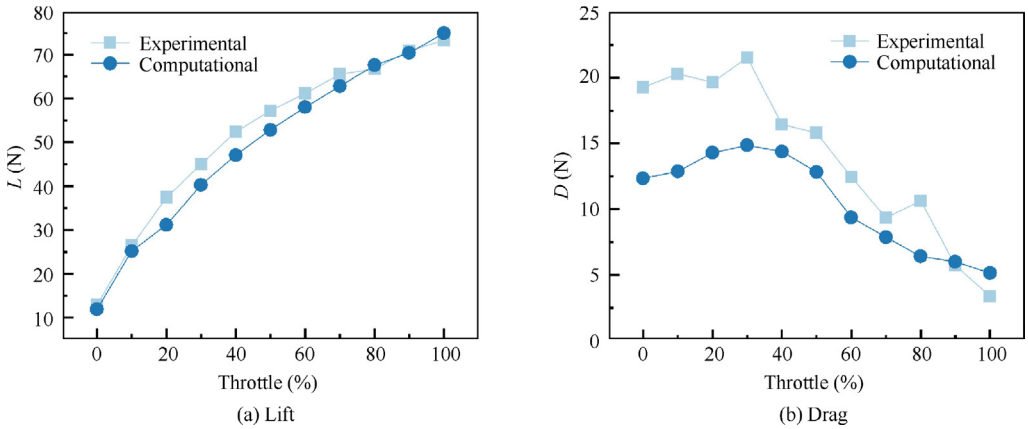



**Fig. 22** Comparison of CFD results of powered wing to experimental results.

and the increased drag and lift loss is nonnegligible, which contributes to the non-command nose-up.

A numerical method is verified by comparing the simulation result to the experimental result. Due to the complex nonlinear flow of the EDF and the simplified CFD model to reduce the expense of simulation, the calculated lift agrees well with the experimental data. However, the accuracy is lacking for calculated drag with a well-predicted trend. We consider the numerical method as a reliable rapid evaluation method for the aerodynamic performance assessment in preliminary design.

As for the deflectable thruster mounting surface, it can also significantly improve aerodynamic performance. With $\delta_T = 30°$, the max $C_L$ can be increased to 2.28, which is 1.6 times the cruising configuration. With the deflection of the surface, the $C_D$ increment caused by the thrust device at a small angle of attack was eliminated, especially for the situation of throttle being less than 20%. Overall, deflection of the power unit led to an increase in thrust and stream-normal force. The increasing thrust of the ducted fans seemed to enhance this effect and produce an increment of lift coefficient for a given thrust deflection angle.

In conclusion, the DEP enhanced the aerodynamic characteristic, and the aero-propulsion coupling was strong. Further study is needed to utilize such a phenomenon to improve DEP aircraft design and optimization.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgement

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